

Side-by-Side Comparison: Deterministic SHE-Boltzmann MOSFET Reproduction vs. Liang *et al.* (1997)

Companion to report Revision 5. Bias $V_{gs} = V_{ds} = 3$ V (Figs. 2–7); converged Anderson 80×66 fixed point, $V_{th} = 0.614$ V.

August 19, 2026

The left/upper panels are reproduced from W. Liang, N. Goldsman, I. Mayergoyz, P. J. Oldiges, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997), for scholarly side-by-side comparison; the right/lower panels are the present reproduction. Every substitution (reconstructed doping, surrogate impact-ionization rate, DD-proxy I – V) is disclosed in the main report; comparisons are structural (peak magnitude, location, profile), not pixel registration.

Fig. 3 — Electron concentration $n(x, y)$

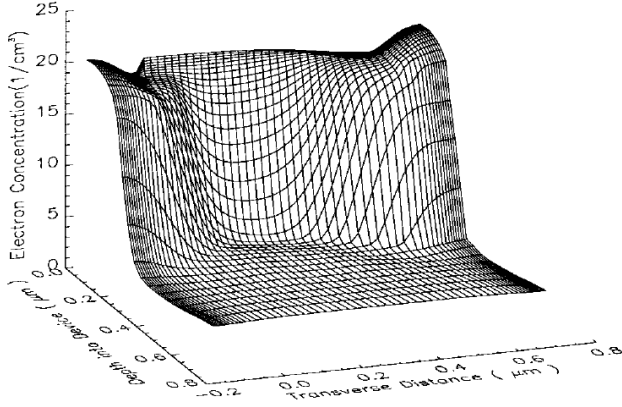
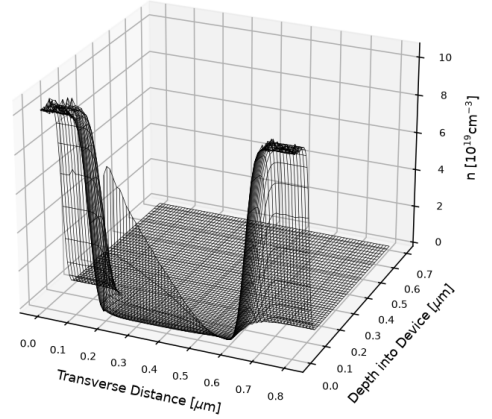


Fig. 3. The calculated electron concentration for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The electron concentration was obtained by integrating the calculated distribution function.

Liang *et al.* (1997)

This work: electron concentration (cf. Liang Fig. 3)



This work

Comparison. Both show $n \sim 10^{20} \text{ cm}^{-3}$ walls at source and drain with a deep channel dip and the onset of drain pinch-off. Our peak $n_{\max} = 1.05 \times 10^{20}$; the wireframe uses 10^{19} units (peak ≈ 10).

Fig. 4 — Average electron velocity

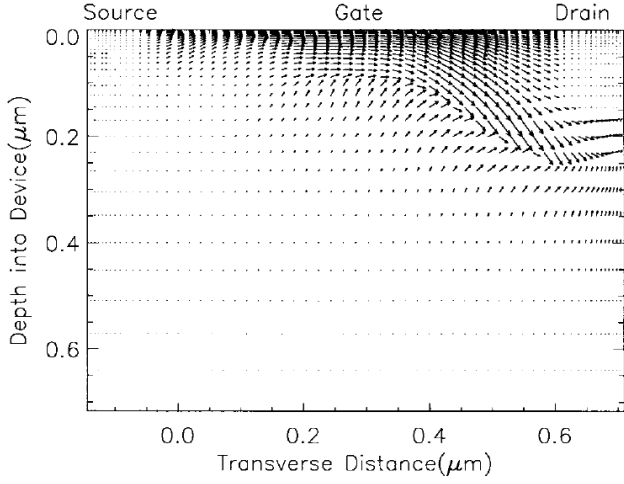
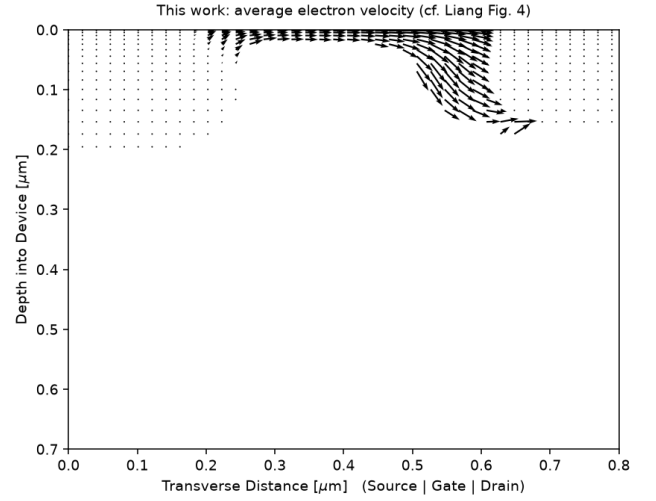


Fig. 4. The calculated electron average velocity for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The current density was obtained by taking the

Liang *et al.* (1997)

Comparison. The flow is lateral through the channel and turns downward into the drain; magnitude saturates at $|v|_{\max} = 1.07 \times 10^7$ cm/s near the drain (Liang $\sim 10^7$). Vectors are masked where $n < 10^{13}$ cm $^{-3}$.



This work

Fig. 5 — Electrostatic potential $\phi(x, y)$

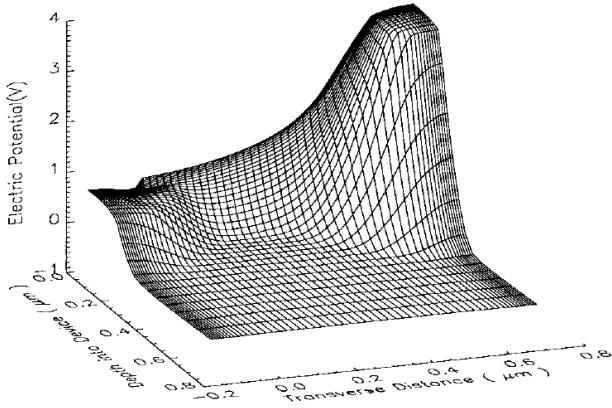
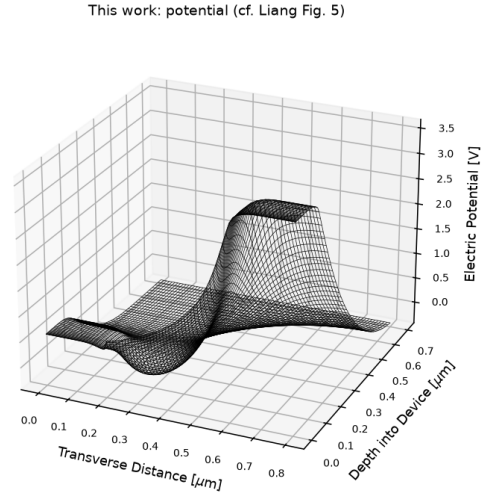


Fig. 5. The calculated electrostatic potential for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The potential was obtained by the self-consistent solution of the Poisson, Boltzmann, and Hole-Continuity equations.

Liang *et al.* (1997)

Comparison. The potential rises monotonically from the source ($\approx$ built-in 0.59 V) to the drain (3.59 V = V_{ds} +built-in), matching the paper's smooth 2-D ramp.



This work

Fig. 6 — Hole concentration $p(x, y)$

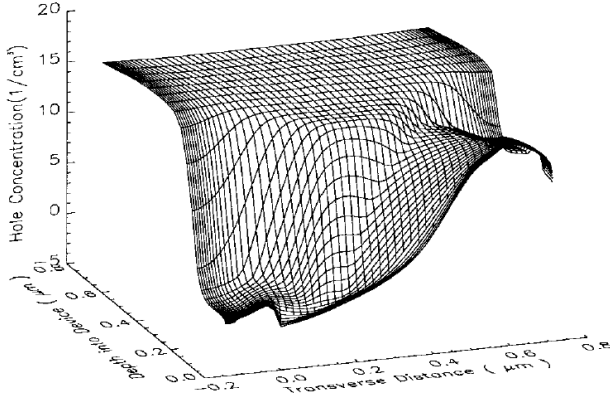
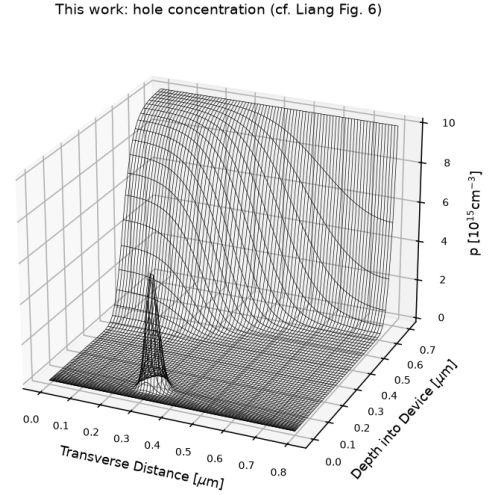


Fig. 6. The calculated hole concentration for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The hole concentration was obtained by the self-consistent solution of the Poisson, BTE, and Hole-Continuity equations.

Liang *et al.* (1997)

Comparison. Holes fill the p-body ($p \sim 10^{16} \text{ cm}^{-3}$) and are depleted in the n^+ S/D; at the converged fixed point the impact-ionization hole source is small (G_{ii} low). Absolute positions depend on the reconstructed doping, so the agreement is structural.



This work

Fig. 7 — Electron temperature (dashed) and G_{ii} (solid) contours

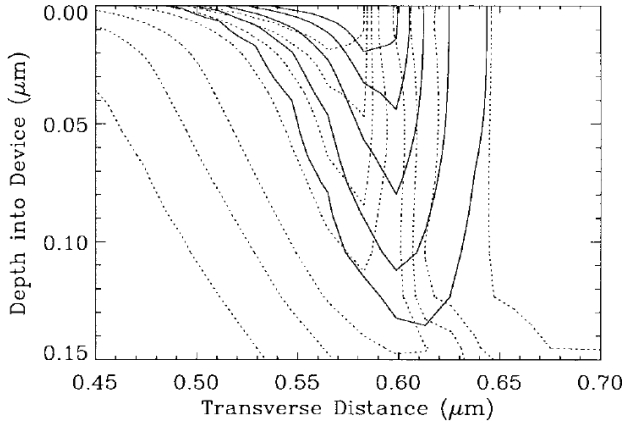
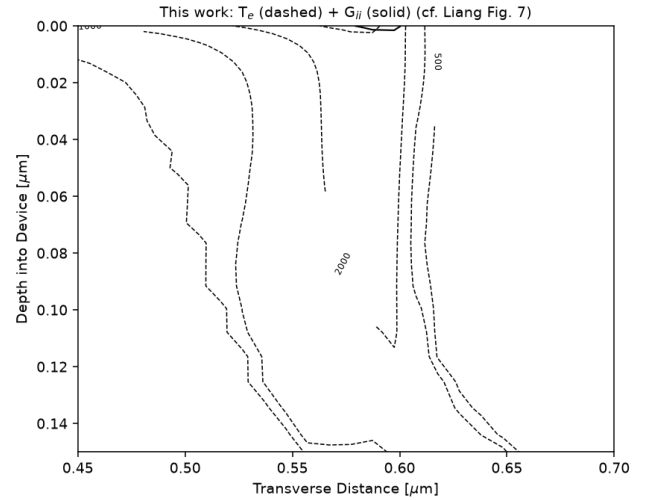


Fig. 7. Contour plots of the electron temperature (dashed lines) and the electron-hole pair generation rate due to impact ionization (solid lines). The contour values for electron temperature are 5000 K, 4800 K, 4500 K, 4000 K, 2000 K, 1000 K, and 500 K. The contour values for electron-hole pair generation rate are $2.3 \times 10^{27} [1/\text{cm}^3 \cdot \text{s}]$, $1.7 \times 10^{27} [1/\text{cm}^3 \cdot \text{s}]$, $1.0 \times 10^{27} [1/\text{cm}^3 \cdot \text{s}]$, $4.0 \times 10^{26} [1/\text{cm}^3 \cdot \text{s}]$, and $1.0 \times 10^{26} [1/\text{cm}^3 \cdot \text{s}]$.

Liang *et al.* (1997)



This work

Comparison. On Liang's own drain-region axes and contour levels (converged Anderson 80×66 fixed point): our T_e reaches 3201 K (64% of the top 5000 K contour) and G_{ii} 1.5×10^{26} (7% of the top 2.3×10^{27}), both drain-localized with T_e far broader than G_{ii} . With the coupling loop now converged (Anderson acceleration), G_{ii} is loop-converged but still *mesh*-unconverged (the $40 \times 34/60 \times 50/80 \times 66$ sequence gives $8.1/2.2/1.5 \times 10^{26}$); the converged value is $\sim 15 \times$ below Liang's — a *model* deficit (surrogate Keldysh II rate on reconstructed doping), not a numerical artefact. The T_e/G_{ii} peak offset stays within one cell.

Fig. 2 — Carrier energy distribution $\log_{10} f_0$ (three depths)

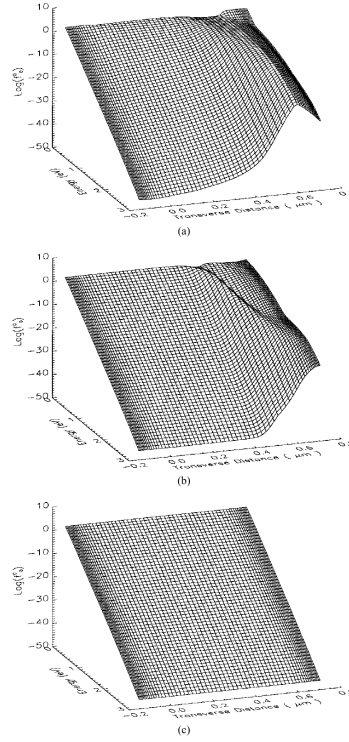
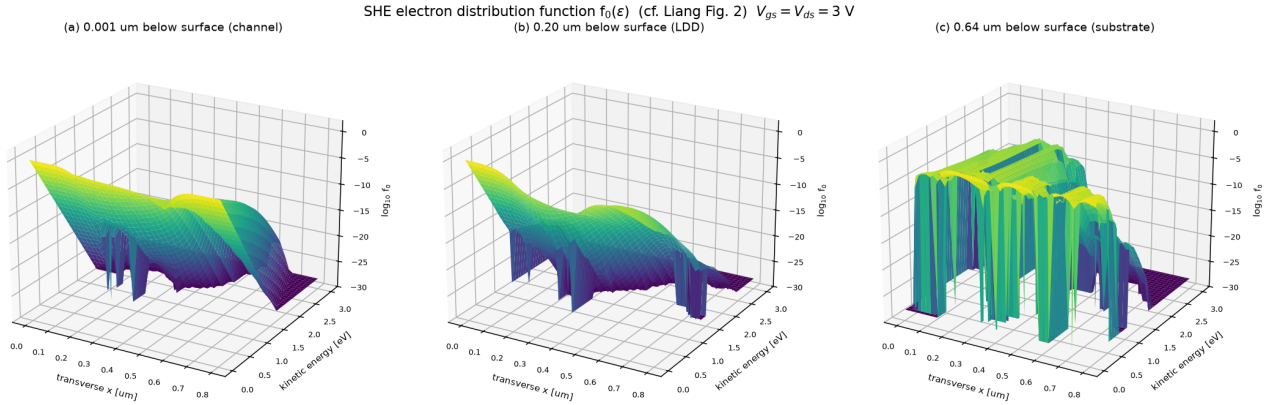


Fig. 2. (a) The distribution function along the channel $0.001 \mu\text{m}$ below the surface. (b) The distribution function along a line $0.2 \mu\text{m}$ below the surface. (c) The distribution function along a line $0.64 \mu\text{m}$ below the surface, which is in the substrate. The applied bias is $V_{GS} = 3 \text{ V}$ and $V_{DS} = 3 \text{ V}$.

In addition to the details of the distribution function, the SH method readily gives accurate values of moments $\langle A \rangle = \int A f_0 d\epsilon$.

Liang *et al.* (1997): (a) $0.001 \mu\text{m}$, (b) $0.2 \mu\text{m}$, (c) $0.64 \mu\text{m}$ below the surface.



This work (same three depths).

Comparison. Near the surface (a) and in the LDD region (b) the distribution develops a hot, non-Maxwellian high-energy tail toward the drain; deep in the substrate (c) it is thermal. Our substrate slice (c) is noisier because f_0 there sits near the numerical floor, but the Maxwellian plane is recovered under the noise.

Fig. 8 — Terminal I–V characteristics

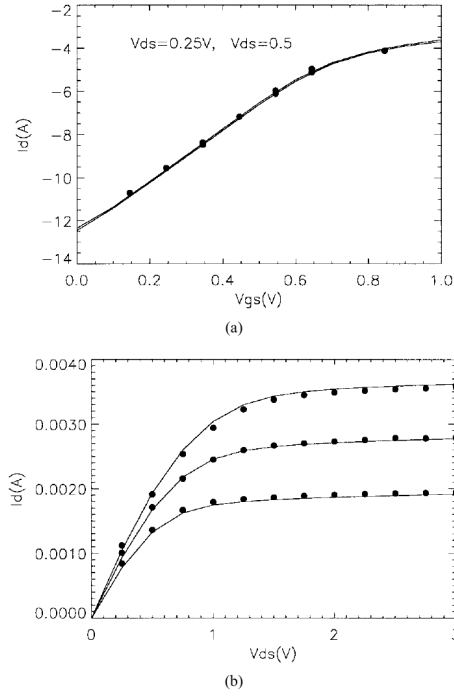
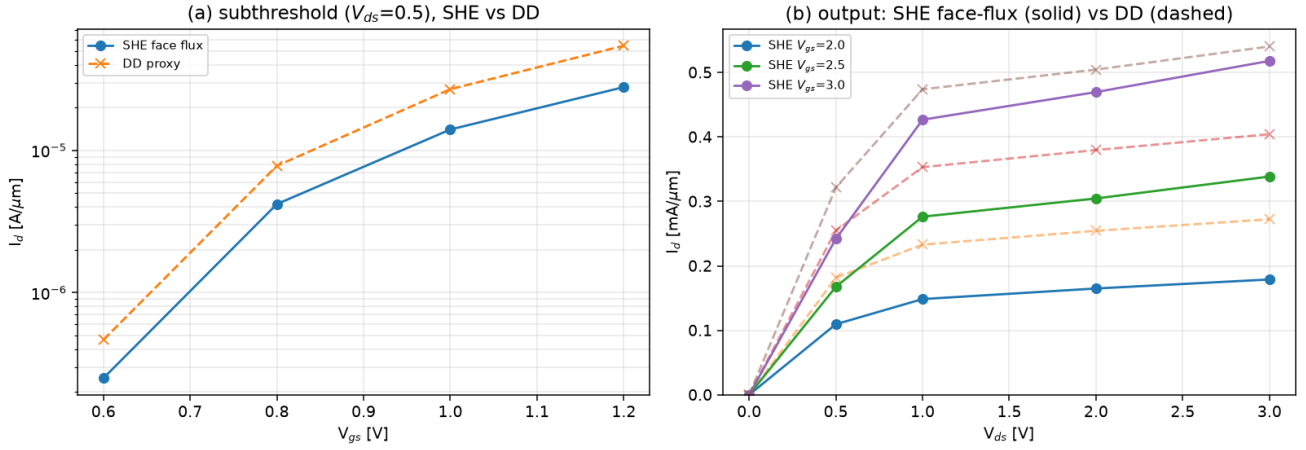


Fig. 8. Calculated and experimental I–V characteristics for the device for a wide range of applied biases. The dots are calculated by our simulator, whereas the solid lines are experimental values. (a) Agreement with experiment in the subthreshold region for a wide range of gate-source voltages with $V_{DS} = 0.25$ V and 0.5 V. (b) Agreement with experiment in the triode and saturation regions for a wide range of drain-source voltages with $V_{GS} = 2.0$ V, 2.5 V, and 3.0 V.

Liang *et al.* (1997): dots = simulator, solid lines = experiment.

Terminal I-V from the SHE velocity moment (Liang's method) vs the DD proxy -- recalibrated device, 40×34



This work: terminal current from the *SHE velocity moment* (finite-volume face flux, solid) vs the DD proxy (dashed), per μm .

Comparison. Like Liang, the terminal current is obtained from the SHE velocity moment (the finite-volume face flux), *not* a mobility model, reproducing the triode→saturation output family and the monotonic V_{gs} ordering; source/drain current conservation is 2×10^{-5} – 10^{-3} . The SHE current runs $I_{d,\text{SHE}}/I_{d,\text{DD}} = 0.51$ – 0.96 (bias-dependent), systematically below the DD proxy retained (dashed) for comparison. This is a *one-way* SHE (on the DD potential), and the subthreshold panel is at $V_{ds} = 0.5$ V only — a partial reproduction of Liang's 0.25/0.5-V Fig. 8(a) family. Liang's Fig. 8 solid lines are moreover experimental data for a real device of unstated width, so only the per-width *shape* is comparable, not the absolute ampere magnitude.